



Partners In Crime

Google and Facebook have reportedly agreed to team up and fight any upcoming antitrust lawsuits. <http://dgit.in/jan21-71>



MuZero

Google's DeepMind division develops an AI that can learn both, chess and Atari classics. <http://dgit.in/jan21-72>

ably identify individuals even when face masks are worn by using both facial and iris recognition technologies simultaneously. Once the Bio-IDiom system detects the presence of a mask in a face image, the engine elects to focus on areas surrounding



NEC's Bio-IDiom system can allegedly identify individuals even with masks on

an individual's eye to extract and analyse individualistic characteristics. According to the company, their system also supports masks of various colours as well as patterns.

Another example is Thales Cogent FRP (Face Recognition Platform), a video-based facial recognition system, which can detect when a person is wearing a face mask and allegedly obtains a recognition accuracy of approximately 99 per cent as long as the reference photo fed into the system is high quality. Thales achieved this by training its deep neural network algorithms with hordes of photos of people wearing masks, so the algorithms are trained to focus primarily on the parts that stay exposed even with masks on.

Photographs on masked individuals are pivotal in training existing facial recognition algorithms to accurately verify a face even with a mask obstructing it. On that note, researchers at Wuhan University in China compiled and posted to Github a dataset of 5,000 cleaned and labelled images of the masked faces of 525 individuals, alongside images of 90,000 unmasked faces. As time passes, more

and more masked images should be added to the dataset which will dramatically improve how algorithms adapt to the new normal. Additionally, simulated masked images that are being created by companies such as NIST provide a more immediate solution to the shortcomings of present-day face recognition technologies.

Closer to home, Pune-based deep tech startup, EDGENeural.ai has claimed that its FacEDGE facial recognition technology works both with and without masks, according to a report by APEX Media. The company's co-founder Sarvesh Devi told APEX that they have modified their system's deep-learned-based facial recognition algorithm as well as their hardware to rise up to the challenges brought on by face masks for facial ID systems. According to Devi, their patent-pending mask-based facial recognition technology boasts of up to 95 per cent accuracy.

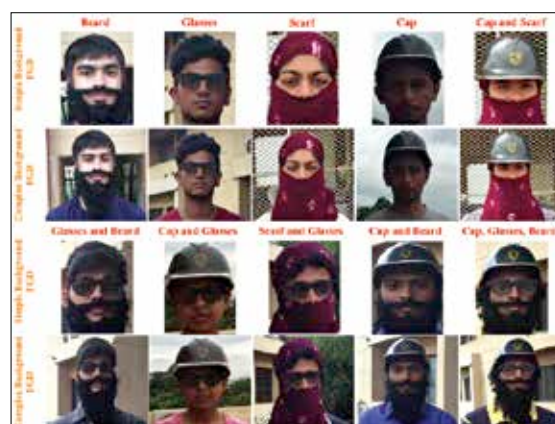
CONTINUED EFFORT

While the pandemic has certainly accelerated the development of facial recognition systems targeted at mask-wearers, the technology has been in development in the academic circles for a while now. Researchers from Stanford University published a study on Disguised Facial Identification (DFI) back in 2017. The facial recognition system developed by this team was more advanced than most commercially available ones since it could identify faces by using fewer key points (facial features and their metrics) than others. A team of Indian researchers including Amarjot Singh (Department of Engineering, University of Cambridge) and his colleagues: Devendra Patil and G. Meghana Reddy (National Institute of Technology, India) and S.N. Omkar (Indian

Institute of Science) also presented their framework to identify people who wear masks or disguises back in 2017 in the paper titled "Disguised Face Identification (DFI) with Facial KeyPoints using Spatial Fusion Convolutional Network", which detected 14 facial key-points to identify even those with face coverings by leveraging deep convolutional networks.

A team from the University of Bradford was able to achieve 90 per cent accuracy with their facial recognition model. In a study, this group demonstrated that they could train a program to accurately identify masked faces by deleting certain parts of the photos that they utilised to train the software. Leveraging AI, the team achieved 100 per cent recognition rates for both three-quarter and half faces.

Facial recognition, before the pandemic, was slowly but surely becoming an essential part of the security infrastructure of airports, banks, buildings, and even smartphones. Efforts that started before the pandemic as well as the development that came after the pandemic



Indian researchers developed a technology which can identify people with various kinds of facial disguises

are paving the way back for facial recognition systems even in the new normal. While face masks and facial recognition systems have shown issues coexisting, the development in the algorithms enabled by large datasets of masked individuals is proving to be a viable solution for this. **4**